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Design Document

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Project Name: Guide Lines

1. [Redacted For Privacy]
2. Stin Dang
3. [Redacted For Privacy]
4. [Redacted For Privacy]

1. Introduction

1.1. Purpose and Scope

Guide Lines is designed with the goals of well-being and usability in times of crisis or in high stress situations that are suddenly thrust upon the user. The use case of our application will be for those who have suddenly been put in a situation where their housing situation, food access, and/or mental health is in jeopardy. The user base we are targeting will have access to a smartphone with no further requirements other than a grasp of English to operate the application.

1.2. Organizational Context

We are targeting organizations that serve a coordinating role across our local crisis management resources, who can help to ensure that they get into the right hands, at the right times. Their guiding values for the project would be to:

- Reduce decision-making stress during crisis using cognitive offloading.
- Enable faster connections to verified emergency resources.
- Ensure inclusivity for seniors, newcomers, and people with impairments.
- Build a feeling of trust through ethically sourced communication that feels well researched and personally relevant.

We also hope to contribute to Sustainable Development Goal #3: “Ensure healthy lives and promote well-being for all at all ages.” This can be achieved by designing our app with the idea of well-being at its core and making use of the existing array of helpful resources that people may not be aware of.

1.3. Background Information

In crisis situations like housing insecurity, food insecurity or mental health struggles, people often don’t have a plan on where to go for immediate help. Existing services are fragmented and not always easy to find in high stress conditions. Our proposed interactive application provides users with a defined pathway for these emergency situations that will assist them in navigating their situation in a respectful and efficient manner.

Crises can happen at any time, and are not necessarily associated directly with an “incident” (Harake, 2024, p. 3). They are often more unpredictable and associated with instability. The existing format of searching for resources can be difficult to navigate in these scenarios because of the different mindsets that an individual can be in. Two of the main phases include aggression, and depression, impacting the individual's ability to commit and follow through on information heavy digital resources (Brandstätter, Herrmann, & Schüler, 2013, para. 3). Our application would have long-term use in the real world, specifically by helping vulnerable peoples with creating a bridge between the resources they need and how to find or access them. Through this app, we see the social good that can come from interacting with it by centralizing how users discover and connect with social support resources.

1.4. Target population

Guide Lines users are adults within a crisis who do not have the normal capacity, time, or experience to plan through an unpredictable situation. They face heightened stress and a desire to quickly find support. Those who engage with our application may have a minimal understanding of the situation they are going through, and lack the medical terminology that can be found in some of the emergency resources. Alternative stakeholders that may use our application include caregivers, especially for the elderly who require assistance in using digital tools to access resources, and community groups that coordinate their actions with other resource providers.

1.5. User Problem Statement

Individuals in crisis situations can experience a variety of cognitive impacts that can inform the design of our application. The mindset of users within these situations is strongly focused on their own past experiences and how the crisis may impact what is going on in the lives of their own or their family (Sato, 2024). This results in a reflective, inward focus on the situation, which can risk users feeling defeated from the crisis’ impact on their aspirations or progress in life. Another potential problem is that users exhibit reduced engagement with resources if the output of content is not balanced. Resources need to be memorable and applicable to the user, however, excessive repetitive content can cause desensitization of the important messaging (Acconito, Angioletti, & Balconi, 2025, p. 500). A balance is necessary to communicate the messaging in the most effective way possible. To address this, our application is aiming to have a consistent list of trustworthy

organizations that are consistently available in these crises, allowing users to have a lifeline in times of uncertainty.

1.6. Initial development approach

Our project is represented by a prototype created in Figma which allows us to visualize the testing of different interaction, flows information architecture, and visual hierarchy. Utilizing Figma as a prototyping platform helps prioritize the design to be functional first within a contained environment without expending the higher time cost of testing designs within a coding environment, greatly increasing the time we can spend on usability and appealing design early, which is important for an app that is designed for a high-stress time in many users' lives. Additionally, to gain a better understanding of our users during development, we outlined a system diagram (figure 1) to identify the pathways that users can take during the use of the application.

2. Design Process

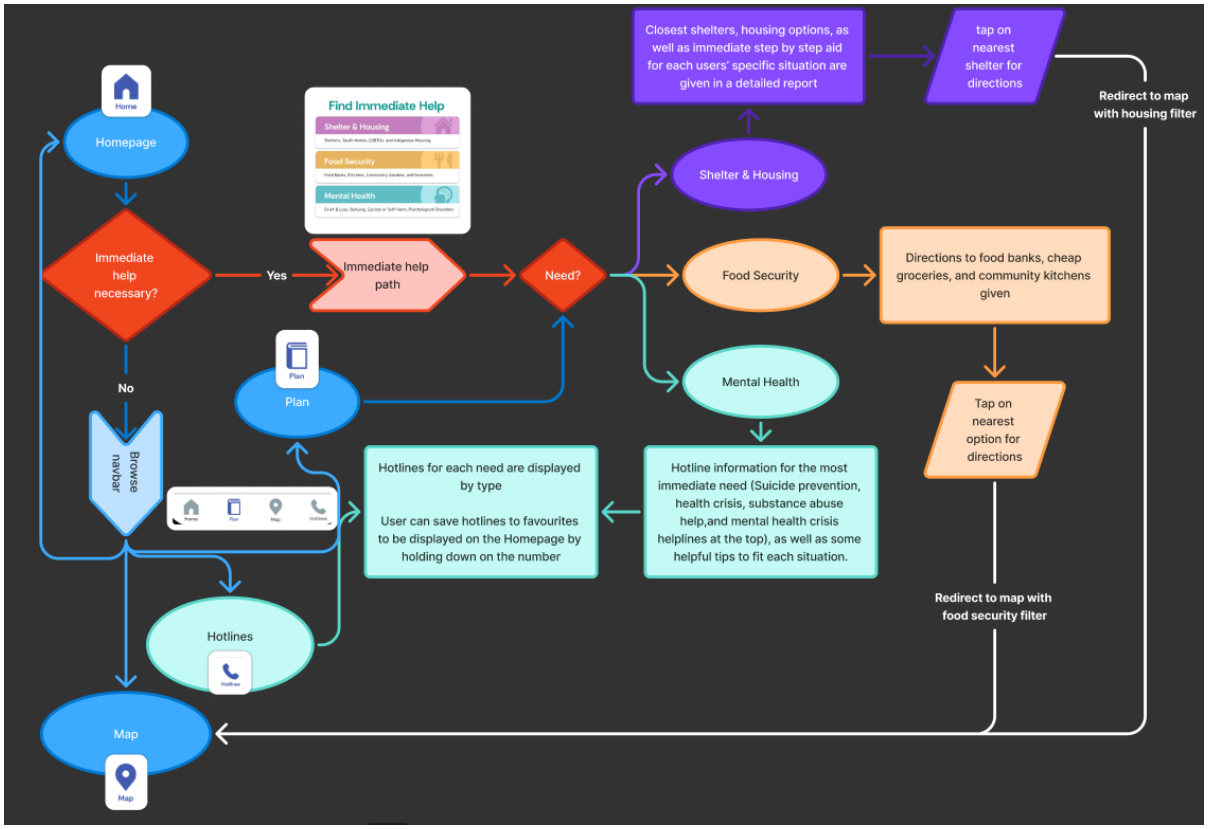


Figure 1: High level system diagram.

2.1. Initial questions about design

Our initial questions when beginning work on design often revolved around what people do during high stress situations, such as “How does elevated stress influence our ability to make decisions?”, “When in times of high stress, how is our cognition affected negatively, and how can we circumvent that limitation?” These questions led us to researching other, similar apps. The apps we reviewed in our research were Wysa, Stay alive, myPlan, and Safety Plan Mobile App. We evaluated them based on the above questions, as well as compared their efficacy in an HCI context, utilizing the three waves of HCI as a framework for evaluation. We found that these apps utilized very streamlined user flows to achieve very quick results in a pinch by clear visuals, minimized input steps, and single action operations that improve the apps usability. In terms of the apps’ design, they often utilized very clean interfaces with little clutter, creating apps that prioritized the users’ needs above trying to impress them visually, which is important for those who need clear information in already stressful times.

For example, myPlan (figure 2) uses a structure that offers each piece of content room to be interpreted and slows the pace of using the app down for better understanding of the important components. The law of proximity also ensures that when a new section begins, additional space is included to denote the change. Similarly, the app Stay Alive shows us a clear relationship between text and the arrows to the right (figure 3). Considering the principle of Fitt’s Law, the arrow icons are efficient at communicating their purpose and close to the right side for faster hand interaction time.

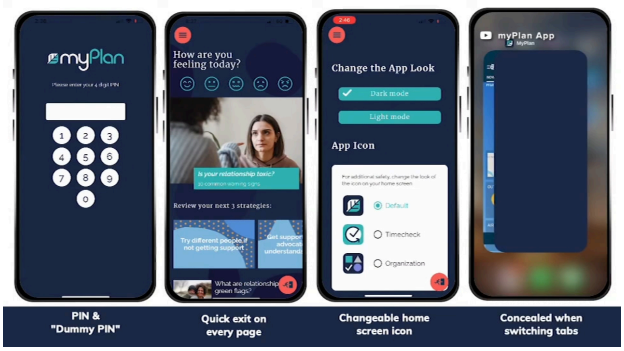


Figure 2: Interface of myPlan app.

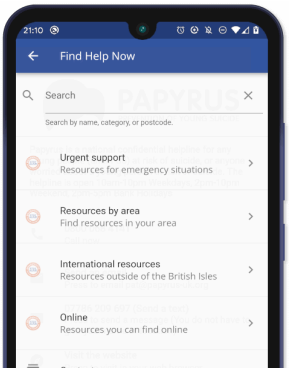


Figure 3: Interface of Stay Alive app.

These apps helped to inform us about balancing large amounts of content and encouraged us to consider what aspects would be obvious right away, and what can be hidden for later discovery.

We took the aspects from these apps that felt intuitive, functional, and fast, and created a homepage that contains a dashboard of different categories, as well as favourite numbers for the user to quickly reference in a pinch. Well defined colours and simple vector icons allow the user, at a glance, to navigate to where they need to go without the cognitive load necessary for reading sentences. This would also have the added benefit of helping those who have a harder time reading English, such as new immigrants or those with reading comprehension disabilities.

2.2. HCI perspectives used to address design questions

Our application designs for a multiwave approach, with primary focuses on user experience and cognitive performance. After conducting research on the types of resources that are available, we noticed that there are many different options that users can take to fit their needs due to the large amount of independent services that are not linked together.

For the content that our resources provide, we followed a GOMS structured method, beginning by identifying the primary user goals in high stress scenarios and the key pieces of information that they would look for as actionable takeaways. Key information includes data that helps users to know if they can make a decision, such as hours, any restrictions like age or gender, and if drop-ins are possible. Users gain context by accessing this data first through the map, to offer a visual overview of the location as addresses can be difficult to memorize with text alone. Then, they can view a details box which summarizes this data into a single pop-up interface, as seen in figure 4.

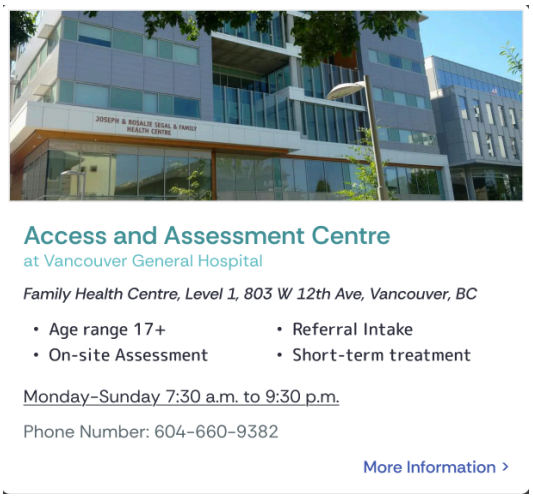


Figure 4: Example of a resource details box.

This details box is important to help the users feel like they can perceive the data from a more manageable perspective because it is applicable to immediate use and concise. Clicking on more information reveals a secondary batch of options that are not as essential and can wait until after the user feels like they know what to do first. This is to reduce the demand on their working memory, notable for crisis resources which often have lots of written information of varying priorities. They can first focus on processing and acting on the relevant information to their situation.

With this process of finding essential information, we recognized that users will have different selection rules for their goals and navigating a map may not fit every user, such as those who just want immediate help like a quick phone call. On the landing page, under find immediate help, the same information is structured in a faster way based on current location. Users can click into one of the three categories to see their easily accessible options with the same decision-making data, including phone number. This is the immediate help path referenced in our system diagram.

Socially, Guide Lines functions as a bridge for users to have easier access to professionals that can best handle their specific circumstances. Based on our research, as mentioned earlier, one of the guiding issues specific to social communication is the balance of using repetition to relay messaging (Acconito, Angioletti, & Balconi, 2025). According to their research, an overuse of repetition in key messages can desensitize users and reduce their willingness to engage in the content. However, it does aim to reinforce important concepts. To still have users recognize the important concepts, we chose colour as a signifier to divide the three most prevalent avenues of crisis resource content. Colour-coded categories that line up with these avenues allow the user to recognize related ideas as one branch of the apps function, creating an association with each colour that meets their intended goal. For example, a user who needs mental health assistance will understand quickly that they are searching for things in the blue category with the mind icon, creating a quick association between screens while interacting with the application. The use of pre-conscious visual processing with these categories is apparent on the landing page where users can correlate similarly coloured content with one another. By using colour-based signifiers, we retain the benefits of users recognizing key ideas through repetition while minimizing the desensitization that can come with repeated text. This creates a clearer, organized design language.

When designing for user experience, we prioritized the role of the app in connection with the user's environment in order to best integrate it into their lifestyle. Beginning with the user, there are many feelings that can be felt during a crisis like distress and hopelessness, and it can be difficult to “think straight” to know exactly what your needs are (Crisis Centre BC, 2022). Considering this state of mind, we identified that using a map as a visual representation for the layout of the resources is the strongest way to communicate a sense of reassurance. We want the user to know that there are solutions and with the map, they can easily correlate their location with nearby resources. Users can navigate a map with just a simple goal in mind, such as “I need to find a shelter in the quickest amount of time”. A map also connects with the users in a way that feels seamless to how they already understand their surroundings. Our society builds on the standard conventions of a paper map, as well as google maps, which provide examples of widely shared mental models that we wanted to build on.

The map, as well as the overall app also considers strategies to apply external cognition, improving the condition of the user’s likely overwhelmed mindset. In order to take advantage of our app as an external piece of media that can simplify cognitive load, features like “recently called hotline numbers” place the user in a position to recognize valuable information as they are working through the crisis management process, rather than having to recall what they previously used. This helps organize the process and removes the pressure of memorizing steps while they’re actively managing the situation. This feature is especially important if they require dealing with multiple hotlines or sources that they are speaking to.

2.3. Design and development constraints

When developing Guide Lines, the question of security was paramount, as many people may feel ashamed of their situations or feel inclined to anonymity for safety purposes. As well, the ethical and technical concerns surrounding how organizations are labelled and displayed in the app were among our first considerations. The way we labelled organizations and subcategorized them was a very selective process, which means it is a very labour and research intensive process. Automating this process would be a very difficult or nigh impossible thing to accomplish, so we had a few ideas of “Opt-in” for a manual review for organizations to partner with the application to ensure that the application does not cross any ethical or legal boundaries surrounding endorsement for

organizations. For example, if we were to partner with an organization that has been known in the past to mistreat its clientele, we would have an ethical problem to tackle by removing the service from our platform's system. Vetting for these programs by researching them beforehand is then necessary for our system to ensure the safety of its users as well as establish credibility and trust in the platform.

Another potential constraint would be with hardware: Phones typically require elevated permissions for location data and network connectivity to interact with the mapping API, creating a potential security vulnerability if our mapping systems are compromised. Despite this, our system does not store any personal data that would identify or compromise our users in any way if our mapping systems were to have their data compromised, apart from location data from the phone. We designed our application with this constraint in mind, ensuring that no user data is saved within the app, ensuring that no login details or identifying information was given to the application for a sign up process. Sidestepping a sign in process with identification did limit the features we were able to offer, but we feel this lightweight approach would ensure security was preserved better, which is something that users of the application would value.

Another concern for the app would be if a sudden influx of users caused a stress onto the hotlines or the surrounding facilities. It would be beyond our control if an influx of users caused these organizations to cease collaborating with the platform for fear of inability to serve so many users. This is a concern that we cannot address apart from localizing the service to larger organizations that can handle an influx of clients, which is something that goes against the purpose of the app, serving as a local resource for people to discover new lifelines in times of crisis, so we cannot address this issue fully without collaborating with other organizations to plan for these types of situations. This also presupposes that our application has a very large userbase, which is a slight possibility, but seems unlikely given the app's use case is very specific to times of crisis.

3. Final Design Description

3.1. System Functions

The main functionality of Guide Lines is to be a repository of information, as well as a guide for those who require assistance in navigating complex and traumatic experiences. The app will serve as both a map and source of relevant information for users that experience food security, mental health, or housing related issues. These three categories are colour-coded within the application to show the user, without needing to read, what sort of resources are near them on the map.



Figure 5: The three colour-coded immediate help pathways.

As well, the application will also serve as a hub for users to view hotline numbers of relevant organizations that operate within their regions, with each number being categorized under each colour coding.



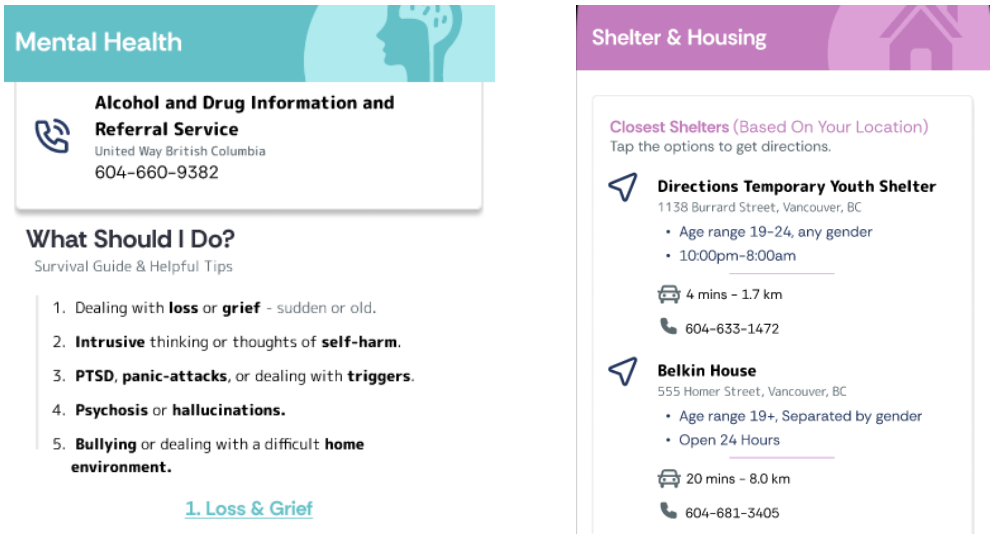
Figure 6: Recently Called Numbers section.

The app tracks the users' most recently used hotlines for quicker use on the homepage, as well as giving them the option to seek more immediate help on the homepage. If users are not in need of that immediate help, they are also able to explore the app, view hotlines in case of emergency, and find other features using our app's navigation bar, located at the bottom of the application.



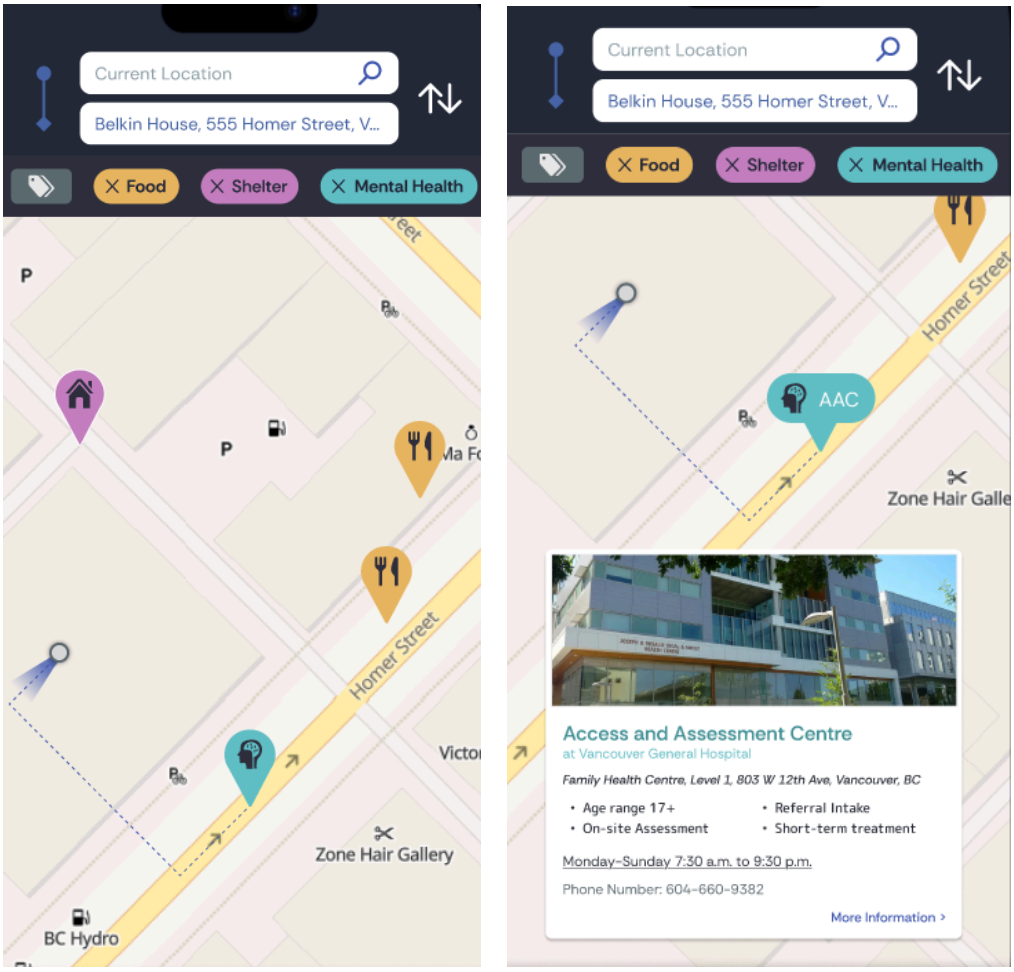
Figure 7: Navigation bar.

Guide Lines' plan function goes into details for each of our three categories, allowing users to get helpful advice and tips for navigating tough times. These tips are categorized according to common issues that users may be facing, highlighted for ease of readability and numbered for quicker navigation.



Figures 8 & 9: Category details.

The map function will allow users to see what services are around them, creating a valuable guide for when people need to seek out help from the resources available nearby. It features filter tags for specific qualities common within the array of resources we collected, as well as tags for the three main categories.



Figures 10 & 11: Map function.

Finally, users navigating to the Hotlines section will be greeted by a list of relevant hotlines that they can call at any time, sorted by their use cases and colour coded according to our aforementioned categories.



Figures 12 & 13: Hotlines section for two of the categories.

Our hotline categories ensure that users are getting help tailored to their specific needs, breaking down and subdividing the issues of “where to turn to” when in a high stress situation.

3.2. Similar System Information

Our application serves as a hub for other systems on the phone, namely the calling application and network permissions. Our functionality depends on being able to access the phone’s calling, allowing for the user to quickly swap between the apps by tapping the call buttons in our hotlines section and home screen. The map functionality also depends on location services provided by the phone’s GPS features, as well as depending on the network connectivity to fetch the list of available services that are nearby to the users, relying heavily on the map API for finding relevant places for the user.

4. Description of Future Work

For further prototyping: User testing utilizing focus groups on the current build would be done to ensure our app is straightforward and usable enough to go forward with our current design with. Another focus group will be taken at a later time that will include those who have dealt with

high-stress incidents – such as what our app is targeting – before. When this user feedback is taken into account and adjusted for, the next step would be to begin building the app through Android Studio, using Java, as it has multiple libraries that are useful for creating the various features we have prototyped, such as map APIs, permissions for phone call intents, and network connectivity. The app would be tested on a variety of android devices for the sake of compatibility, and then ported to an iOS environment using Xcode after the app has largely finished iteration. For compatibility reasons, the android version would use Java 17, which can support a large variety of older devices, something that ensures the app is usable for those who may not have the newest or most powerful phone. This ensures stability and usability throughout the range of devices, as well as having the added benefit of being a very fast java version, creating a quicker user experience in the end. The bulk of the work would be to create a categorization feature using the map, finding and categorizing the different services by location and displaying them correctly will take a lot of database indexing and correct labelling to ensure that users are not misled by the map feature, as many places need to be listed under the app's three categories, and accuracy in that regard matters greatly for our users.

To get the app into the hands of as many people as possible, a website to serve as an information repository and overview of the app would be created after release to promote the application. The website would function similarly to other product websites, such as the website for the app discussed prior, Wysa, which would serve to promote trust in the product from potential users, establishing credibility for us as the creators. The website would also include resources for the application, links to the Google Play Store and Apple App Store, as well as a press kit for journalists to review or write on the application.

> Link to Figma Prototype <

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